

RITESH MAMIDI

AI / ML Engineer • Product & Data Analyst | Data Engineering • Generative AI • MLOps • Product Analytics

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PROFESSIONAL SUMMARY

AI/ML engineer and data/product analyst who builds systems end to end — from large-scale **Spark & Kafka data pipelines** to **LLM evaluation, RAG, and MLOps**, and the **product analytics** that turn model output into decisions. 4+ years across data engineering, machine learning, AI evaluation, and analytics. Currently drive data-quality and AI-evaluation work on **Apple Maps** for a product used by hundreds of millions. Author of 20+ production-shaped open-source projects. MS in Business Analytics; M.Tech in Software Engineering; IBM AI Engineering certified.

CORE COMPETENCIES

Data Engineering: Apache Spark / PySpark, Spark SQL, Structured Streaming, Apache Kafka, Cassandra, ETL/ELT, dimensional modeling, medallion lakehouse, Parquet, data-quality frameworks

AI / ML & LLMs: PyTorch, scikit-learn, TensorFlow, LLM evaluation (LLM-as-judge), RAG, AI agents, NLP, computer vision, prompt engineering

MLOps & Serving: FastAPI, Docker, model registry & stage promotion, drift monitoring (PSI/KS), feature stores, CI / pytest

Analytics & BI: Product analytics, A/B testing & experimentation (CUPED, guardrails), funnel & retention, forecasting, KPI design, Power BI, Tableau

Data & Cloud: SQL (PostgreSQL, MySQL, Snowflake, BigQuery), Python, R, AWS, Azure, GCP, Git/GitHub, Linux

PROFESSIONAL EXPERIENCE

Data & AI Analyst — Apple Maps

Sep 2025 – Present

Apple (client engagement) • Austin, TX

- Design and maintain **automated data pipelines and data-quality frameworks** unifying model-quality, reviewer, and policy data into production reporting.
- Evaluate **LLM & ML model outputs** against published policy and quality rubrics; co-led a model-evaluation initiative surfacing systematic failure modes.
- Ship Power BI dashboards and Python/LLM prototypes; cut operational escalations **30%** and improved investigation efficiency **18%** with QC frameworks.

Business / Data Analyst

Jan 2024 – Aug 2025

Heart Hospitality LLC • United States

- Built **ETL pipelines and automated reporting** (Python, SQL) across business systems — **55%** less manual reporting effort and real-time visibility.
- Designed relational data models and predictive forecasting/classification models, improving planning accuracy **22%**.

Data / Business Analyst

Jan 2022 – Dec 2023

Liberated Monks Pvt. Ltd. • India

- Built Python/SQL ETL pipelines and Tableau dashboards informing acquisition, engagement, and retention.
- Built ML segmentation and propensity models improving marketing-driven engagement **18%**; owned data validation and KPI definitions.

SELECTED PROJECTS (OPEN-SOURCE • RITESHMAMIDI0905-LAB.GITHUB.IO)

Real-Time Streaming Pipeline — Kafka → Spark Structured Streaming → Cassandra fraud detection with event-time windows, watermarks, and a one-command Dockerized stack.

Spark Data Lakehouse — end-to-end PySpark medallion (bronze→silver→gold) lakehouse on 300K+ events with broadcast-join enrichment, partitioned Parquet marts, AQE tuning, and data-quality checks.

LLM Evaluation Framework — rubric-driven platform scoring LLM/agent outputs on faithfulness, hallucination & refusal via LLM-as-judge, with a REST API and interactive dashboard.

MLOps Platform — feature store, experiment tracking, model registry with stage promotion, FastAPI serving, and PSI/KS drift monitoring.

EDUCATION & CERTIFICATIONS

Master of Science, Business Analytics — St. Francis College, United States • **M.Tech (Integrated), Software Engineering** — Vellore Institute of Technology (VIT), India

IBM AI Engineering — Professional Certificate (Coursera) • **Google Advanced Data Analytics** — Professional Certificate (Coursera)